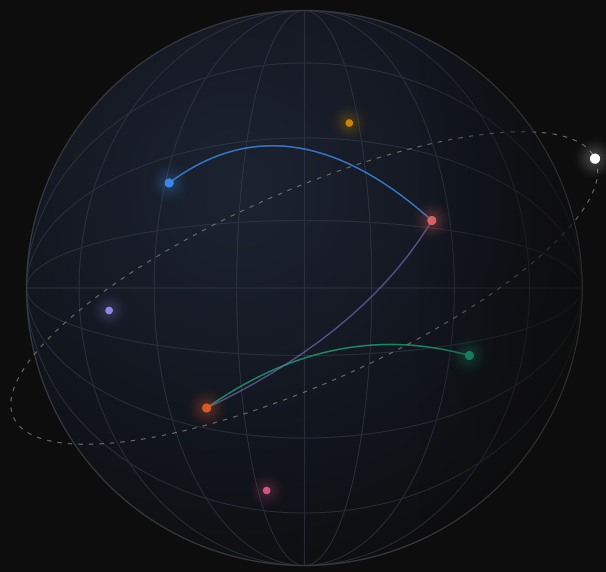

GSOC MONITOR

ONE GLOBE · EVERY SIGNAL

A real-time 3D operations console for Global Security Operations — live weather, wildfire, seismic, and infrastructure hazards, a tracked fleet of aircraft, ships, and satellites, open-source intelligence, and a full incident-command suite, fused onto a single CesiumJS globe.

25+
live map layers, each toggleable with its own controls

46
external data feeds — government, satellite, community & commercial

29
server proxy routes with caching & graceful degradation

39K
lines of TypeScript — React + CesiumJS + Node/Express + Postgres

01 · THE PLATFORM

A mission-control shell around a living globe

GSOC Monitor renders a sun-lit, atmosphere-shaded Earth on a procedural starfield and lets an operator stack live data on top of it — from live weather to police scanners. Everything is real-time, everything is clickable, and every layer explains where its data comes from.

45 EXTERNAL FEEDS
NOAA · NWS · USGS · NASA FIRMS · NIFC · LANDFIRE · CelesTrak · adsb.fi · AISStream · Blitzortung · GDELT · NPS · Bluesky · PulsePoint · Socrata · 18 utility outage feeds…

→

NODE/EXPRESS + POSTGRES
29 proxy routes with TTL caches that serve last-known-good data when upstreams fail; background collectors for lightning, ships, rivers, wind & OSINT; keys held server-side.

→

REACT + CESIUMJS CLIENT
25+ layers on a 3D globe, dockable panels, dashboards, crisis tools, and screensaver tours — adaptive from ops wall display down to a phone.

BASE MAP · WEATHER · FIRE & SMOKE · HAZARDS & ALERTS · OPEN-SOURCE INTEL · TRACKING · LOCATIONS · CRISIS OPS · DASHBOARD · SCREENSAVER

Click anything, get a panel

Every feature on the globe — a quake, an alert polygon, a ship, a satellite, a river gauge — opens a draggable, resizable detail panel; 20 panel kinds, color-accented by category, with snap-to-dock zones. Where several features overlap, a picker lists everything under the cursor.

Layer console with live status

A sectioned sidebar toggles every layer and shows a live one-line status straight from its store — "412 strikes/min · live", "3 active systems", "7/7 ships tracked" — plus inline controls: magnitude chips, satellite groups, flood-stage filters, proximity radii. All state persists across reloads.

Provenance built in

An "i" button documents every active layer: who provides the data, how it is gathered, and why it can be trusted, with a canonical link — USGS to Blitzortung. The trust story travels with the map.

Kiosk-grade self-recovery

Built for unattended wall displays: dual GPU watchdogs detect WebGL context loss or silent frame stalls, classify the GPU, rebuild the viewer up to 4 times behind a "Restarting graphics…" overlay, and auto-reload with a bounded budget so broken hardware can't loop forever.

Four basemaps. Dark, Light, a custom night-tuned "Dark Sat" satellite mode, and topographic — with place labels that fade smoothly by camera altitude.

Re-engineered camera. Zoom, rotate, and tilt split into distinct gestures; a wheel notch feels identical at street scale and globe scale; bounded 150 m–45,000 km.

Global search. Address, place, or raw lat/lon — geocoded through a cached server proxy, flying to a point or framing a region.

Live HUD. Clock, "deployed X ago" build stamp, and an all-layers tracked-item counter with a rolling sparkline.

3D buildings & terrain. OSM building footprints with real heights over Cesium World Terrain, height-gradient styled for the dark palette.

Time zones. ~120 UTC-offset-tinted zone polygons; click any zone for a live-ticking world clock.

Mobile-adaptive. Panels become a swipeable bottom sheet, the sidebar an off-canvas drawer; safe-area aware, PWA-ready.

Performance-tiered. A one-time GPU probe classifies software/virtual/integrated GPUs and tunes resolution, antialiasing, and terrain detail to match.

UNDER THE HOOD · Cesium request-render mode — frames only on change, near-zero idle GPU · one global drill-pick handler across 25+ layers · zustand stores with versioned localStorage migrations · Vite-baked deploy timestamp

The full live-weather picture

An interactive global wind field you can probe with the mouse, every active NWS alert as a polygon, dual-source air quality, smoke plumes, forecast rainfall, and ~12,700 river flood gauges — each backed by a named feed with its own refresh cadence.

Live global wind particles

7,000 particles advect through a bilinearly-interpolated GFS wind grid at ~30 fps, dragging fading trails and colored by speed — trade winds, westerlies, and the jet read as flowing streamlines. Hydrates instantly from a local cache; refreshes every 30 minutes.

Wind probe, pins & point forecasts

Hover anywhere for a live arrow and compass HUD (mph/kt/m/s); right-click drops persistent wind pins; click a pin for a 48-hour hourly wind-and-gust chart with direction arrows plus a 7-day outlook, computed in the location's own timezone.

Every NWS alert, resolved to real shapes

All active National Weather Service alerts refresh every 60 s, colored semantically by hazard family with severity draw-order. Storm-based warnings use their precise polygons; county-based alerts that ship without geometry are resolved to county shapes via FIPS codes. Click for the full official text.

12,700 river gauges with hydrographs

NOAA NWPS gauges in seven flood tiers with severity filters and a "forecast to worsen" highlight. Click a gauge for stage vs. flood thresholds, forecast crest, a 14-day interactive hydrograph, NWS impact statements, and record crests.

Dual-source air quality on one scale

Official EPA AirNow monitors render as numbered AQI badges beside thousands of PurpleAir community sensors — raw PM2.5 corrected server-side with the EPA Barkjohn equation and 2024 AQI breakpoints so both sources agree. Click any station for pollutant-by-pollutant detail.

Wind streamlines. A camera-adaptive field of glowing flow lines with true-tangent arrowheads — density and length re-grid at every zoom, from full globe to street level.

Smoke plumes. NOAA HMS analyst-drawn polygons in three densities, with observing satellite and observation window per plume.

Forecast rainfall. NOAA WPC QPF accumulation overlay (24 h to 5-day windows) with the exact 18-stop official color ramp.

Never blank. The wind layer falls back live feed → Postgres snapshot → baked-in grid, flagging stale data with a HISTORICAL badge.

UNDER THE HOOD · 13 MB river snapshot refreshed off the request path — the API answers instantly · stale-on-error TTL caches degrade outages to slightly-old data, never a 502 · county-geometry fallback for zone alerts · shared refcounted wind-grid loader

03 · SEVERE WEATHER

Lightning & hurricanes

Live global lightning with cinematic strike animations and a replayable 24-hour history, and worldwide tropical-cyclone tracking with no ocean-basin blind spots.

Real lightning, seconds after it strikes

The browser connects straight to the Blitzortung community network: each detected strike plays a jagged bolt descending from 120 km in 150 ms, a red impact flash, then a glowing crosshair that ages white → yellow → orange → red over 10 minutes — up to 2,500 on screen, with a live strikes-per-minute ticker.

24-hour strike history, deploy-proof

Sidebar chips replay the last 1/6/12/24 hours as an age-tinted point cloud of up to 20,000 strikes. Behind it, the server runs a 24/7 collector whose ring buffer is persisted to Postgres in gzip-packed 5-minute chunks — the history survives restarts and deploys.

Hurricanes with true cyclonic spin

Every active tropical cyclone is a rotating three-arm spiral glyph — counter-clockwise north of the equator, clockwise south — tinted by Saffir-Simpson category, with cone of uncertainty, observed and forecast tracks, and hoverable forecast points showing winds in mph and knots.

Global formation watch

NHC 7-day outlook areas render as risk-tinted polygons with formation-odds bars — and for the basins NHC doesn't cover, the server parses JTWC's plain-text advisories into mapped invest areas with Low/Medium/High potential. West Pacific typhoons are not a blind spot.

Efficient at scale. Strike fields idle at zero GPU cost between updates; impact flashes animate alpha only, replacing per-frame re-tessellation.

Storm detail on click. A cyclone's panel shows sustained winds, gusts, minimum central pressure, position, basin, advisory time, and a link to NHC; forecast dots enlarge on hover with a +72h-style tooltip.

Formation summary. The sidebar rolls the world up to one line: "2 active systems · 3 areas to watch."

UNDER THE HOOD · speaks Blitzortung's undocumented wire protocol (LZW frame decoder, relay rotation) · 3M-slot typed-array ring buffer spans 24 h at ~200 strikes/sec · regex parser for JTWC's hard-wrapped bulletins

Five interlocking fire layers

Detection, response, forecast, fuels, and analysis: live satellite hotspots, named incidents with the full response picture, the official 7-day fire-potential outlook, a 30-meter fuel-model raster, and a draw-your-own-zone fire-risk analyzer.

Satellite hotspots with science-grounded assessment

VIIRS thermal detections from the past 24 h, sized by Fire Radiative Power. Clicking one yields a 5-level flaming-vs-smoldering call derived from FRP and I-4 brightness temperature — including explicit "sensor maxed" saturation handling — with confidence, satellite, and pass time in local timezone.

Named fires with the response picture

NIFC/WFIGS interagency incidents as containment-tinted markers with acreage labels and perimeter polygons for fires over 100 acres. Panels show acres, an animated containment bar, assigned personnel, complexity, cause, and an InciWeb link — the same system of record the responding agencies use.

Draw a zone, get a fire-behavior score

Draw a circle or free-form polygon on the globe and the app runs a zonal histogram against the 30 m LANDFIRE raster: exact per-pixel fuel inventory, burnable percentage, and a 0–100 Fire Behavior Potential score with spread-rate and flame-length categories and plain-English drivers, from published Scott & Burgan fire science.

The official 7-day outlook

~234 Predictive Service Areas in the NWCG palette — Critical and Ignition zones over fuel-dryness shading — with a 7-day picker that switches instantly. The server shrinks ~45 MB/day of government polygons to ~270 KB.

LANDFIRE fuels overlay. All 40 Scott & Burgan fuel models at 30 m across CONUS in the official colormap, with a 7-group legend and hover blurbs.

Near-pins filter. Restrict hotspots to 5–200 miles of the tracked properties with haversine precision; the sidebar shows a live count.

What you draw is what you analyze. The same geodesic ring geometry drives both the on-globe outline and the raster query.

Zero-key data paths. FIRMS, WFIGS, and LANDFIRE stream browser-side from open government services — no API keys, no backend load.

Perimeters, biggest first. Up to 500 fire perimeters and 800 incidents of 5+ acres render simultaneously, with acreage labels that auto-hide when zoomed out.

A drawing UX that guides you. Live radius readout while dragging circles, rubber-band polygon preview, dashed close-the-ring hint, Undo/Finish/Esc — and committed zones stay pinned for side-by-side comparison.

Instant day switching. All seven outlook days arrive in one payload, so flipping between them never refetches.

Smoke + fire together. Pairs with the NOAA HMS smoke layer and AQI stations for the full fire-air-quality picture on one screen.

UNDER THE HOOD · exact histogram-bin → fuel-code decode verified against ground truth · Douglas-Peucker polygon simplification server-side · visibility-aware polling (wall display polls, background tab costs nothing) · per-tier SVG billboards cached as data URIs

05 · TRACKING

Flights, ships & satellites

Worldwide tracking of a named aircraft fleet, the seven-ship Windstar cruise fleet, and up to 2,500 satellites — with real orbital mechanics, dead-reckoned ship tracks, and hand-modeled rotating 3D wireframes.

A multi-source fleet position engine

The seven Windstar vessels are tracked by MMSI/IMO through a layered pipeline: a live AIS WebSocket, supplemented by VesselFinder or MyShipTracking APIs, or a satellite-AIS-carrying CruiseMapper fallback — whichever reported most recently wins. Positions and 72-hour breadcrumb histories survive restarts, and a debug endpoint diagnoses exactly which source is delivering.

Real orbital mechanics, in the browser

Space stations, brightest, GPS, weather, and Starlink groups — every position computed by SGP4 propagation of CelesTrak TLEs, with comet trails, correct far-side occlusion, and adaptive tick rates that keep 800+ Starlink satellites smooth.

Click a satellite, trace its orbit

One click draws the full orbital ring — near half in front of the globe, far half occluded, recomputed as Earth rotates — while the panel re-propagates every second: live position, altitude, speed, period, inclination, apogee/perigee, NORAD ID.

Hand-modeled rotating ship wireframes

Each vessel's card features a slowly rotating 3D wireframe of its actual class — the Star-class yacht's terraced stern decks, the Wind-class schooner's raked masts and sails (Wind Surf correctly gets five) — plus class badges and a full specs grid. The same geometry renders in WebGL, on 2D canvas, and full-size on the ocean.

Live aircraft. Tracked tail numbers worldwide via ADS-B, drawn at true altitude and heading; landed jets persist as dimmed last-seen markers instead of vanishing.

Predicted ship tracks. Breadcrumb trails plus a 6-hour dead-reckoned great-circle projection ending in a predicted-position dot.

Fleet roster fly-to. Click any vessel in the sidebar to enable the layer and fly to its live position; favorites, paths, and groups persist.

One-click "Track the ISS." Enables the layer, switches groups, flies to a 5,000 km framing against the curve of the Earth, and opens live telemetry.

Age-aware rendering. Ship icons fade with AIS report age and show an amber "last known position" banner beyond coastal coverage.

Status at a glance. "2 of 3 tails tracked" · "7/7 ships · live feed" · "2,410 satellites · live" — right in the sidebar.

Flight telemetry. Panels show altitude, ground speed, heading, vertical rate, squawk, and ICAO24 — with a star-to-favorite button.

Type-coded ships. Hull icons tint by AIS ship type — cargo cyan, tanker orange, passenger green — rotated to true heading, gold-ringed when favorited.

UNDER THE HOOD · client-side SGP4 for 2,500 satellites (satellite.js) · Cloudflare-aware scraper with three-strategy coordinate parsing · great-circle dead reckoning in spherical trig · one wireframe geometry, three renderers — WebGL, 2D canvas, Cesium polylines

Multi-hazard monitoring around real assets

Thirty company properties across 13 groups — national-park lodges, offices, a cruise fleet, a cog railway — cross-referenced live against earthquakes, fires, weather alerts, and power outages, with one-click emergency routing and exportable incident reports.

18-feed power-outage aggregation

One toggle lights up live outages across ~19 states from three families of public feeds — state-EM ArcGIS services, the KUBRA Storm Center data behind major utilities' own maps (Oncor, Georgia Power, Evergy…), and NISC co-ops. Each outage carries customers affected, elapsed time, restoration ETA ("overdue 30m" turns red), and cause. No API keys; each source fails independently.

One-click emergency briefing

Click a property pin and the app builds a live safety briefing: road distance and drive time to the nearest police and fire stations, A/B/C options for the 3 nearest hospitals and hotels, and turn-by-turn routes drawn on the globe as animated pulse lines via a custom GLSL shader — with Google Maps handoff and share links.

Property watch proximity scanner

Cross-references all 30 properties against FIRMS fire detections, active NWS alerts (true point-in-polygon tests), and USGS earthquakes at a 25/50/100-mile radius — listing only properties with something nearby, ranked worst-first, with countdown-to-expiry alert chips. Pops out into independently-refreshing dockable windows.

Downloadable PNG hazard report

Any property panel exports a polished report card — gradient header, live watch status, color-barred alert cards, fire rows with distance and radiative power, quake rows with magnitude badges, or a green all-clear — rendered on a 2× retina canvas, dated, and ready to paste into an incident channel.

Earthquakes. USGS live feed as seismic-ripple markers with magnitude/period filter chips; panels show depth, felt reports, tsunami flag, and a link to the event page.

Property pins. Canvas-drawn teardrop pins, terrain-clamped so they sit correctly on mountainsides, in 13 group colors.

Measure tools. Click-to-draw geodesic distance and spherical-excess area with dual metric/imperial readouts — real ellipsoid math, not flat-earth approximations.

Nearest sister properties. Road-routed distances to the three closest properties, each with a "Fly there" button.

Live lightning ticker. Global strikes-per-minute with a pulsing LIVE badge and rolling sparkline, inside the property watch widget.

Data-quality curation. Two live-tested outage feeds were excluded for stale phantom records — correctness over raw coverage.

UNDER THE HOOD · reverse-engineered KUBRA quadkey tile descent with loss-proof cluster fallback · Overpass + OSRM routing with progressive 5→280 km search radii · single throttled scanner shared across every open panel · background-rebuild pattern: the outage API always answers instantly

A build-your-own OSINT operation

Police and fire scanners, city crime data, social media, world news, and park alerts — continuously ingested, auto-classified, geolocated, and pinned live onto the globe, extensible by the whole team through a shared watchlist.

Seven source types, one live feed

Every 90 seconds the intel engine fans out across RSS/Atom feeds, Google News topic searches, Bluesky accounts and keyword searches, PulsePoint fire/EMS dispatch, California CHP incidents, and Socrata city crime datasets — normalizing everything into one stream, auto-classified by severity (info/watch/urgent) and category (scanner, crime, crash, fire, weather, news, social).

Team-shared source watchlist

Any signed-in teammate adds sources through a plain-language form — "News topic", "Bluesky keyword", "Scanner (PulsePoint)", "Crime dataset" — and they flow into everyone's feed within ~90 seconds. Sources persist in Postgres, pause/delete per row, and can be pinned to a place so their stories land on the map.

News on the map, tinted by sentiment

The GDELT layer plots worldwide geocoded coverage of security-relevant events — protests, evacuations, shootings, hazmat — grouped into pins that grow with article count and shift indigo → red as average tone drops. Cards show a hero image, sentiment, and article links; a scope control filters to within 100–500 miles of the property pins.

Automatic headline geolocation

Items arriving without coordinates are geocoded server-side from their place text — with a 30-day cache, a strict per-cycle budget honoring Nominatim's 1-request/second policy, and a miss-blacklist so an unresolvable headline never burns budget twice.

Intel pins. Category-tinted teardrops — scanner blue, crime pink, fire red — with urgent items enlarged and warning-ringed so structure fires read at a glance.

Breaking-news widget. BBC, Guardian, Sky, NPR, and Al Jazeera merged, severity-scored, gazetteer-geolocated, with NEW badges and custom user feeds.

Bottom-edge ticker. A severity-dotted marquee of headlines whenever the news panel is closed, toggling between world news and National Park news.

Park intelligence. Official NPS alerts and news for six tracked parks, severity-mapped (Danger → critical), plus official boundary polygons.

Click-to-fly feed. Clicking a geolocated intel item flies the camera to it at 120 km; degraded sources surface as an amber warning strip, not silence.

UNDER THE HOOD · in-Node AES-256-CBC decryption of PulsePoint's encrypted CAD payload · SSRF defense-in-depth on team-added sources · Promise.allSettled fan-out — one dead source never breaks the feed · shared GDELT cache keeps all clients under the 1-req/5s limit

Incident command & the ops dashboard

A full incident-command suite — FEMA ICS org charts, draw-on-the-globe zones, live public share links, and print-ready after-action reports — paired with an auto-scanning property dashboard that scores threats and writes its own shift briefing.

ICS/NIMS org chart with drag-and-drop staffing

Every incident starts with the standard FEMA structure pre-built — 21 roles from Incident Commander through the four General Staff sections. Drag people from a personnel pool onto roles, add custom sub-roles, collapse branches. Doctrine is enforced: one person, one command role at a time, with timestamped reassignment history.

Draw incident zones on the live globe

Fire perimeters, flood zones, staging areas, exclusion zones, and search grids drawn by clicking vertices on the 3D globe with live preview — or imported by pasting coordinates in decimal, DMS, or WKT with automatic lon/lat detection. Finished shapes clamp to terrain with distance-fading labels and get snapshot thumbnails for reports.

Live public share links

One click publishes a read-only situation report at a token URL — no login — that updates itself in real time over Server-Sent Events as the GSOC edits: org chart, action log, and map, ~1.5 s behind live. Personnel contact details are stripped before publishing; revoking a link instantly closes every open viewer.

The dashboard writes its own briefing

The property status board condenses each 75-second scan into a signals payload and has Claude compose a 12-word headline plus a grounded operational narrative — with prompt-injection hardening, a deterministic fallback when no key is configured, and response caching so multiple wall displays share one model call.

Shared incident workspace. Cards with pulsing status dots, live staff/action/layer counts; every edit auto-saves to Postgres on a 1.5 s debounce — the whole GSOC sees one picture.

Actions & events log. An editable chronology with a timeline view; attached photos are browser-compressed and uploaded to Cloudinary, appearing in share pages and reports.

After-action PDF. Stood-down incidents generate a print-ready report: summary, full org chart, complete log with photos, map, and per-layer thumbnails.

Explainable 0–100 threat score. Weighted, human-readable contributions — worst alert severity, fire within 10 miles, outage scale — with the top three reasons in the tooltip.

Live event feed. Only genuinely-new events surface — including tropical systems approaching a property or vessel, and a maritime anomaly flag when a ship's AIS goes silent over 6 hours. Click to fly.

Split-screen editing. The situation report edits beside the live, interactive globe — drawn layers stay visible while you type.

Ten incident types. Wildfire, hurricane, chemical/HazMat, mass-casualty, cyber, active threat, and more — with a three-state status selector and archive lifecycle (admin-gated deletion).

Property status board. One card per property group, re-scanned every 75 seconds: alerts with expiry countdowns, live temperature and wind, forecast rain, AQI, nearest fire, quake, news, and outage — click to fly there.

UNDER THE HOOD · Postgres JSONB incidents with debounced store-diff sync · SSE fan-out with snapshot-on-connect and push revocation · multi-format coordinate parser with per-line errors · single batched Open-Meteo call covers every property group

Self-driving tours, war-room login, resilient core

When nobody is touching it, the globe flies itself: cinematic tours of properties, parks, live hazards, and the ISS — narrated, if you like. Underneath, an invite-only auth layer and a platform engineered to stay up through database and upstream outages.

Pins tour: cinematic property & fleet fly-around

The camera visits every property and live cruise ship in a shuffled queue, performing a slow 32-second orbit around each — properties get a glowing terrain-clamped "loot beam" in their group color; ships get an expanding sonar shockwave, a rotating wireframe card, live hazard rows, and a reverse-geocoded minimap.

ISS live chase cam

The camera locks onto the International Space Station's real position — propagated by SGP4 every frame — and slowly circles at 2,600 km while the Earth wheels past underneath, with live lat/lon, altitude, and NORAD designation updating every second.

A login screen that's a production

A holographic SVG wireframe globe with pulsing meridians, lightning, and satellites riding orbit paths over a twinkling starfield — plus a hidden Asteroids-style ship you can actually fly with the arrow keys, and a "Virtual War Room" status panel with a live downlink sparkline.

Built to stay up

The schema self-migrates at boot inside a transaction and retries with backoff — a database outage degrades only DB-backed routes while the globe stays live. TTL caches serve last-known-good data when upstreams fail; lightning history, wind grids, incidents, and the watchlist all survive dyno restarts in Postgres.

Global tour. Slow rotation punctuated by fly-downs to curated geopolitical landmarks, live M5+ quakes, and active alerts — with breaking-news locations injected mid-tour.

Parks tour. National parks with animated glowing boundaries and rotating trivia; Hover mode orbits any clicked point, terrain-sampled to clear obstructions.

Voice narration. A Web Speech narrator that scores available voices to find the best neural option and announces each stop.

Invite-only signup. Rotating 8-character codes (no ambiguous characters), bcrypt cost-12 passwords, 30-day JWTs in httpOnly cookies; first user becomes admin.

Admin panel. One-click code rotation, full team roster with roles, and member removal — enforced server-side on every route.

One-dyno deploy. Ships to Heroku as a single Node process serving the built client, with Postgres as the only add-on; nearly all feeds are free.

Focus cards while touring. A bottom card tracks each orbited property's live hazards, top cards combine each property group's active alerts, and a corner minimap reverse-geocodes the view.

Chrome that gets out of the way. All UI auto-hides while a tour runs and returns — with your open panels — the moment it stops.

UNDER THE HOOD · per-frame SGP4 for the chase cam · terrain-aware cinematography (25-point obstruction sampling, 500 m clearance) · zero-WebGL login globe (SVG + SMIL) and slippy-tile minimap — no second GL context · port binds before migrations: boot never blocks

Every data source, on the record

All 46 external feeds. The overwhelming majority are free public services; the few keyed or paid feeds degrade gracefully when unconfigured.

SOURCE	PROVIDES	
USGS Earthquakes	Real-time seismic events	free
NWS api.weather.gov	Active alerts with geometry	free
NOAA NWPS	~12,700 river gauges + hydrographs	free
NOAA WPC	Forecast precipitation (QPF)	free
NOAA HMS	Satellite smoke plumes	free
NOAA NHC	Hurricane tracks, cones, outlooks	free
US Navy JTWC	Invests in non-NHC basins	free
Open-Meteo (GFS)	Global wind grid + forecasts	free
EPA AirNow	Official AQI monitors	key
PurpleAir	Community PM2.5 sensors	key
Blitzortung.org	Live lightning strikes	free
NASA FIRMS	VIIRS fire hotspots	free
NIFC / WFIGS	Named fires + perimeters	free
NWCG Predictive Services	7-day fire potential	free
USGS LANDFIRE	30 m fuel-model raster	free
18 utility outage feeds	Per-outage detail, ~19 states	free
adsb.fi	ADS-B aircraft positions	free
AISStream.io	Live AIS ship stream	key
VesselFinder	By-IMO fleet positions	paid
MyShipTracking	Alternate fleet positions	paid
CruiseMapper	Satellite-AIS fallback	free

SOURCE	PROVIDES	
CelesTrak	Satellite TLE element sets	free
GDELT Project	Geocoded global news	free
News RSS × 5+	BBC, Guardian, Sky, NPR, Al Jazeera	free
Google News RSS	Topic-search intel	free
Bluesky API	Social feeds + keyword search	free
PulsePoint	Fire/EMS dispatch (CAD)	free
CHP CAD feed	California highway incidents	free
Socrata portals	City open crime data	free
NPS Data API	Park alerts + news	key
NPS boundaries	Park polygons	free
Esri USA Counties	County shapes by FIPS	free
Nominatim (OSM)	Geocoding	free
Overpass API	Hospitals, police, fire POIs	free
OSRM	Turn-by-turn road routing	free
BigDataCloud	Reverse geocoding	free
Anthropic Claude	AI duty-officer briefing	paid
Cloudinary	Incident photo hosting	key
CARTO basemaps	Dark/light tiles	free
Esri World Imagery	Satellite basemap	free
OpenTopoMap	Topographic basemap	free
Cesium ion	Terrain + OSM buildings	key
Google 3D Tiles	Photorealistic cities (wired)	paid
Natural Earth	Time-zone boundaries	free

EXTENDING · the architecture is built for new layers: add a LayerId, a layer component, a panel registration, a proxy route, and a sidebar toggle — hurricanes, lightning, FIRMS, ships, and satellites all shipped this way after Phase 1